Unsupervised Environment Design

- Based on underspecified environment

Algorithm 1: PAIRED.

Randomly initialize Protagonist π^P, Antagonist π^A, and Adversary $\tilde{\Lambda}$;
while *not converged* **do**

Use adversary to generate environment parameters: $\vec{\theta} \sim \tilde{\Lambda}$. Use to create POMDP $\mathcal{M}_{\vec{\theta}}$.

Collect Protagonist trajectory τ^P in $\mathcal{M}_{\vec{\theta}}$. Compute: $U^{\vec{\theta}}(\pi^P) = \sum_{i=0}^{T} r_t \gamma^t$

Collect Antagonist trajectory τ^A in $\mathcal{M}_{\vec{\theta}}$. Compute: $U^{\vec{\theta}}(\pi^A) = \sum_{i=0}^{T} r_t \gamma^t$

Compute: $\textsc{Regret}^{\vec{\theta}}(\pi^P, \pi^A) = U^{\vec{\theta}}(\pi^A) - U^{\vec{\theta}}(\pi^P)$
Train Protagonist policy π^P with RL update and reward $R(\tau^P) = -\textsc{Regret}$
Train Antagonist policy π^A with RL update and reward $R(\tau^A) = \textsc{Regret}$

Train Adversary policy $\tilde{\Lambda}$ with RL update and reward $R(\tau^{\tilde{\Lambda}}) = \textsc{Regret}$

end

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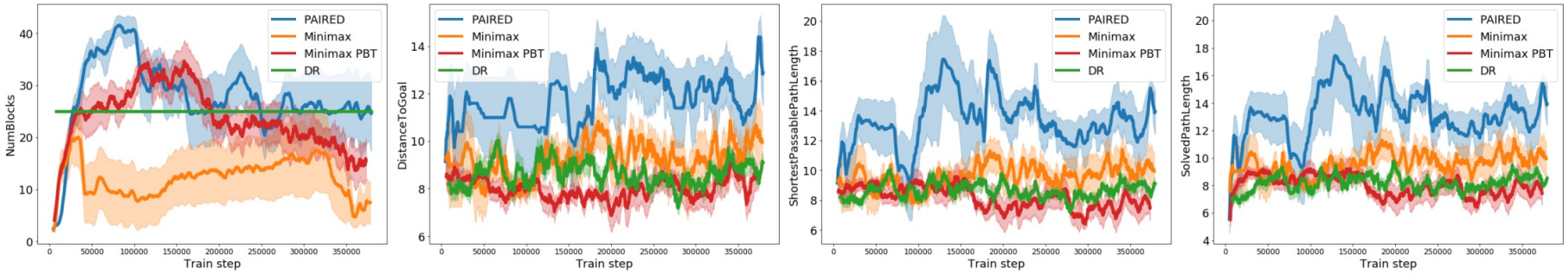
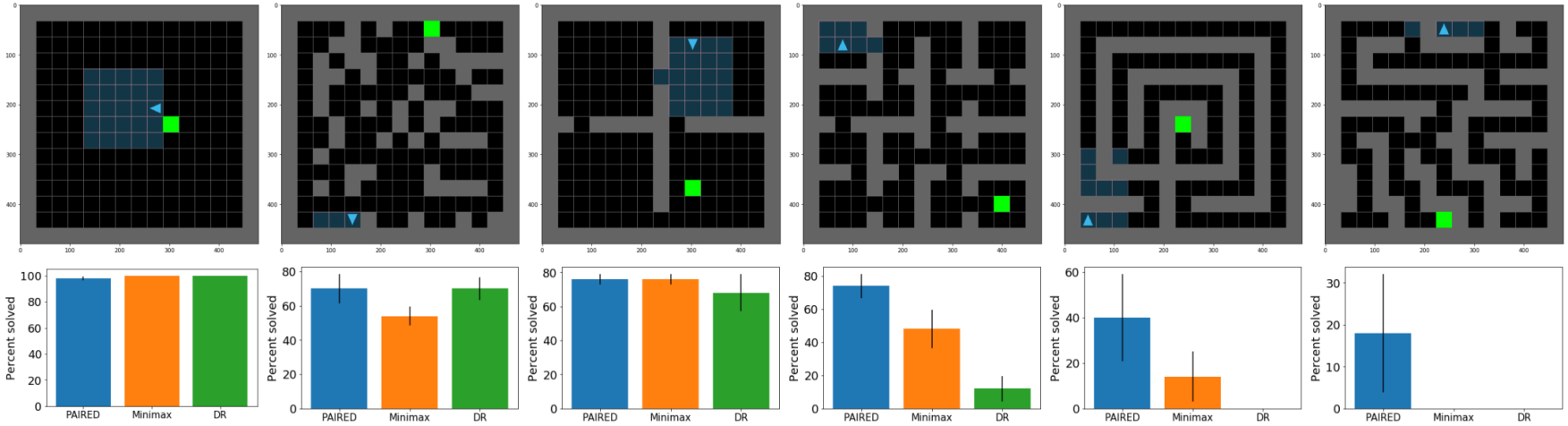


(a) Number of blocks (b) Distance to goal (c) Passable path length (d) Solved path length



(a) Empty (b) 50 Blocks (c) 4 Rooms (d) 16 Rooms (e) Labyrinth (f) Maze